

April, 2026



Well Construction Principle

Lecturer's Profile



Redha Bhawika Putra

❑ EDUCATION

- ❑ **2010** - Bachelor of Electrical Engineering, Institut Teknologi Bandung
- ❑ **2012** - Master of Science of Advance Control & System Engineering, University of Manchester

❑ WORKING EXPERIENCE

- ❑ **2012-2016** Drilling Engineer – Chevron Geothermal & Chevron Pacific Indonesia
- ❑ **2016-2021** Drilling Engineer – Star Energy
- ❑ **2021-2022** Senior Consultant Drilling Engineer – KS Orka Renewables
- ❑ **2022-Present** CEO of Arka Data, Tech Developer for Geothermal and Oil and Gas Drilling
- ❑ **2024-Present** Tenaga Ahli Pengeboran Panas Bumi ESDM

❑ ORGANIZATION

- ❑ **2019-Present** Co Founder & Secretary General of Jakarta Drilling Society (JDS)
 - ❑ Participate in Rig Merah Putih Project initiated by PT Pertamina and PT Pindad
 - ❑ Develop Geothermal Well Drilling National Standard (SNI)
- ❑ **2025 - Present** IATMI, SPE, IA MCR, IAE ITB, JCI Indonesia
- ❑ **2021 - 2024** President of Indonesian Association of British Alumni

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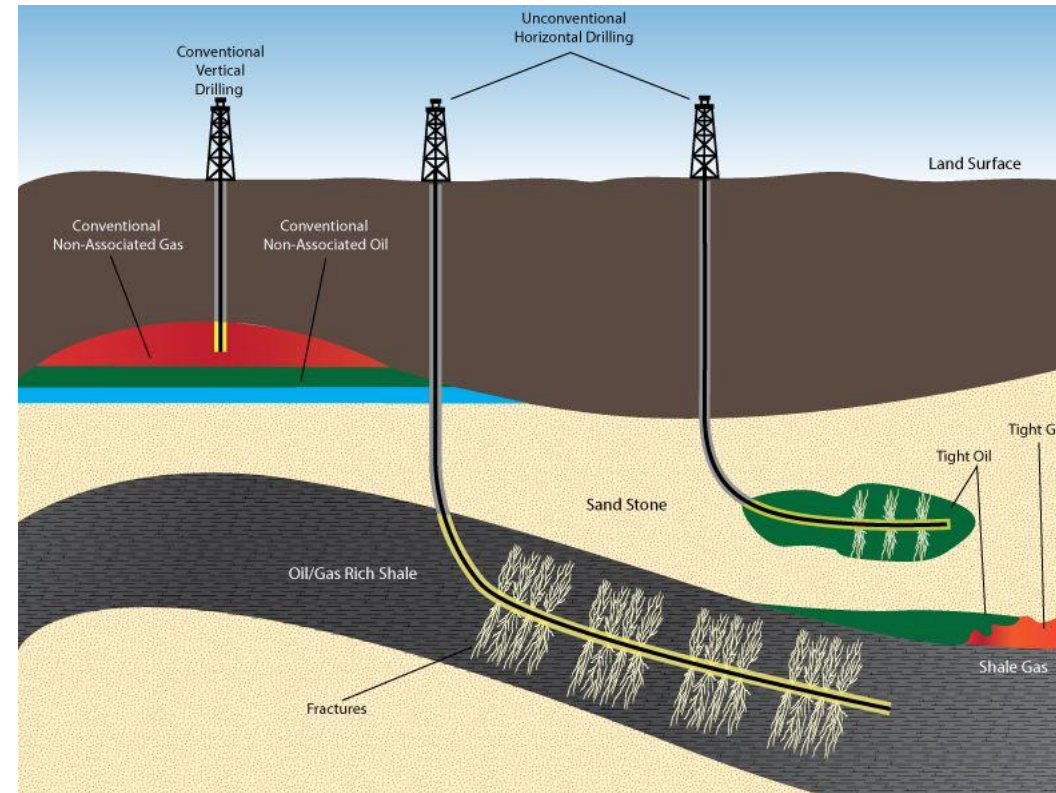
DRILLING INTRODUCTION



Drilling Introduction



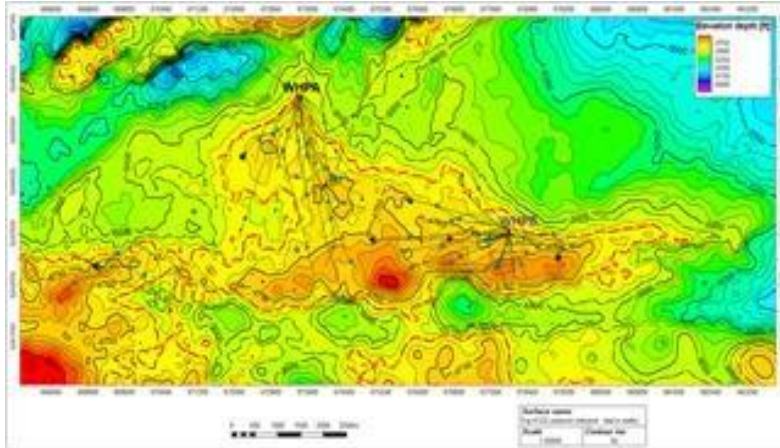
Drilling is the process of boring a well into the earth to reach target reservoirs of oil, gas, or geothermal resources using a drilling rig and supporting systems.



Drilling Project Lifecycle

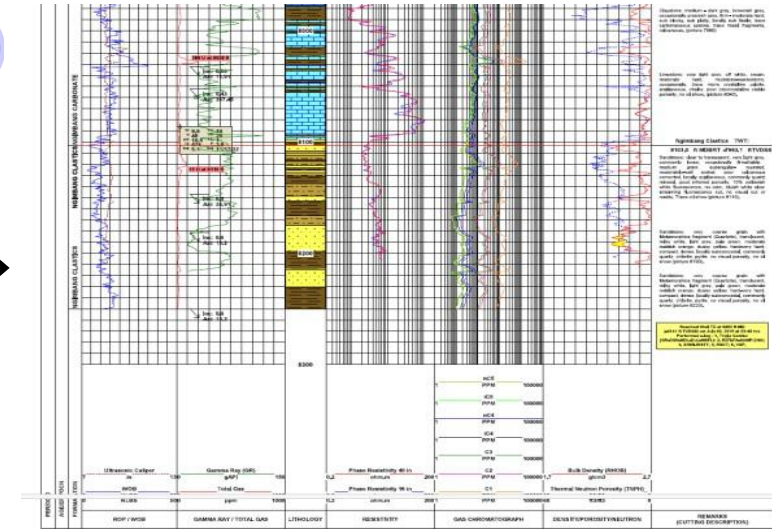
1

Geological Maps



Geologist, Petro-physicist, Geophysicist

2



Reservoir Engineer, Production Technologist

3

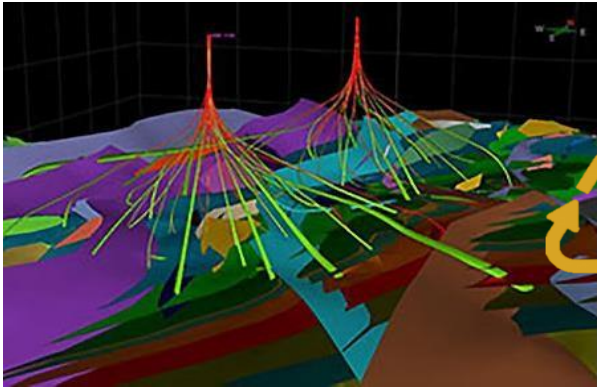


Surface/Facility Engineer

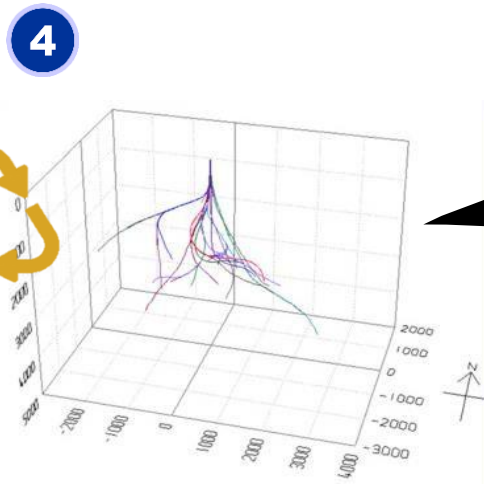


- ✓ Surface Location
- ✓ Bottom-hole Location
- ✓ Offset well information
- ✓ Shallow & Deep hazard
- ✓ Surface (platform)

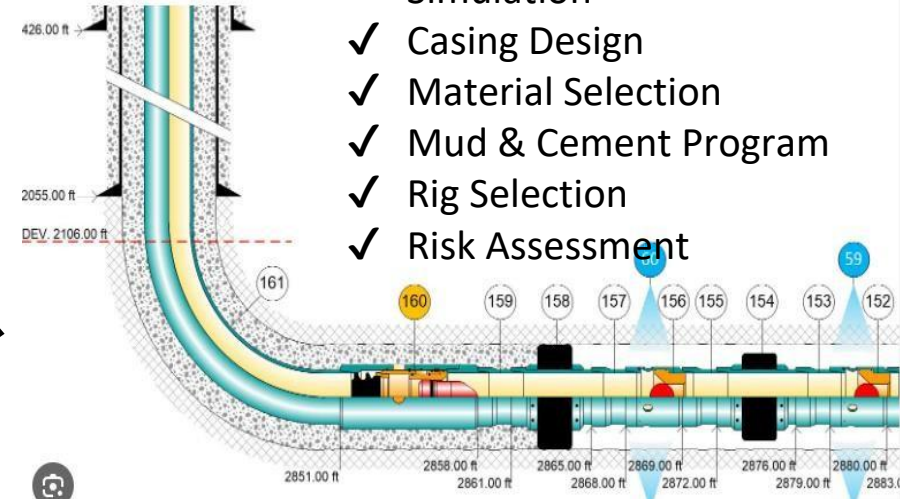
Drilling Project Lifecycle



Well Trajectory



5



- ✓ Torque & Drag Simulation
- ✓ Casing Design
- ✓ Material Selection
- ✓ Mud & Cement Program
- ✓ Rig Selection
- ✓ Risk Assessment



6

- ✓ Time and Cost
- ✓ Economics analysis
- ✓ SKK Migas Approval
 - ✓ Plan of Development
 - ✓ AFE

Execution:

- ✓ Drilling Program
- ✓ Rig Supervision
- ✓ End of Well Report
- ✓ Lesson Learnt
- ✓ Close out report

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WELL CONSTRUCTION PROCESS

WELL DESIGN

Well Trajectory

Well Depth

Well Size

WELL COMPONENTS

Casing Design

Wellhead Design

Cement Design

WELL EXECUTION

Drilling Rig

Drilling Tools

Drilling Fluid

WELL DESIGN



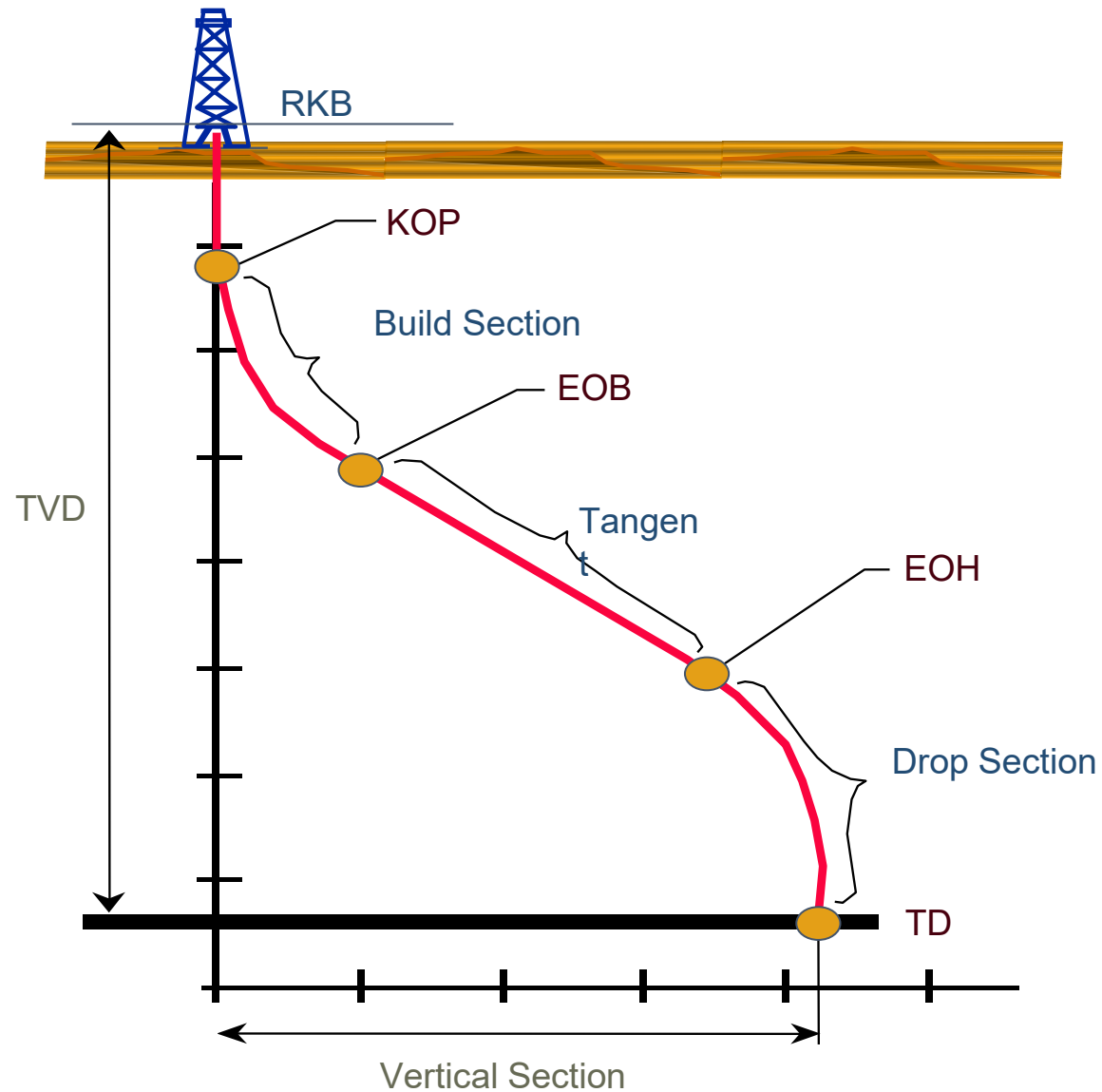
WELL TRAJECTORY

Well trajectory is the path of a wellbore from the surface location to the subsurface target.

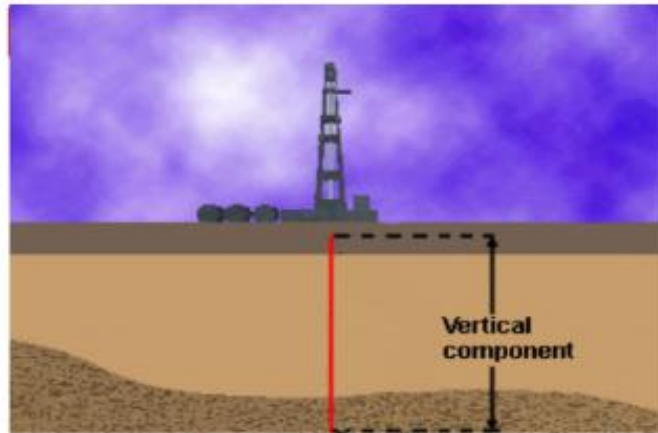
Controls the direction of the well, reach the right place safely and efficiently

Well Trajectory

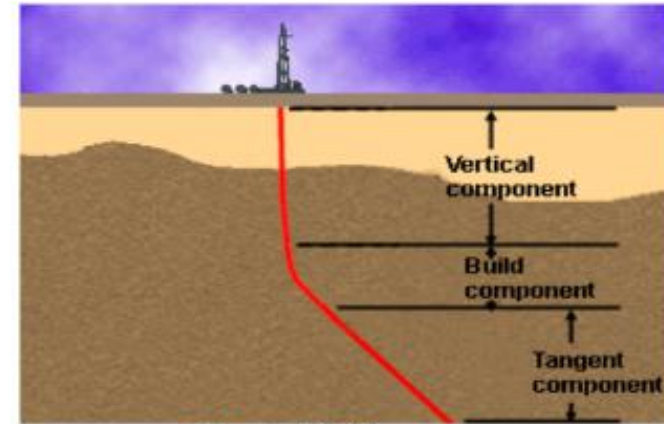
RKB = Rotary Kelly Bushing
KOP = Kick-off Point
EOB = End of Build
EOH = End of Hold
TVD = True Vertical Depth
TD = Well Depth



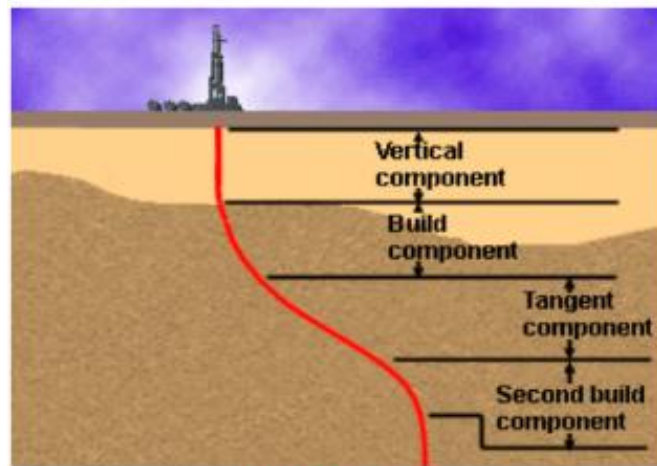
Well Trajectory



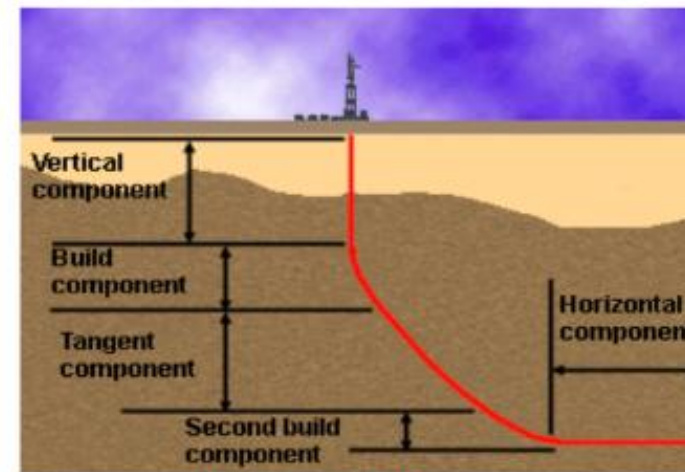
Vertical Type



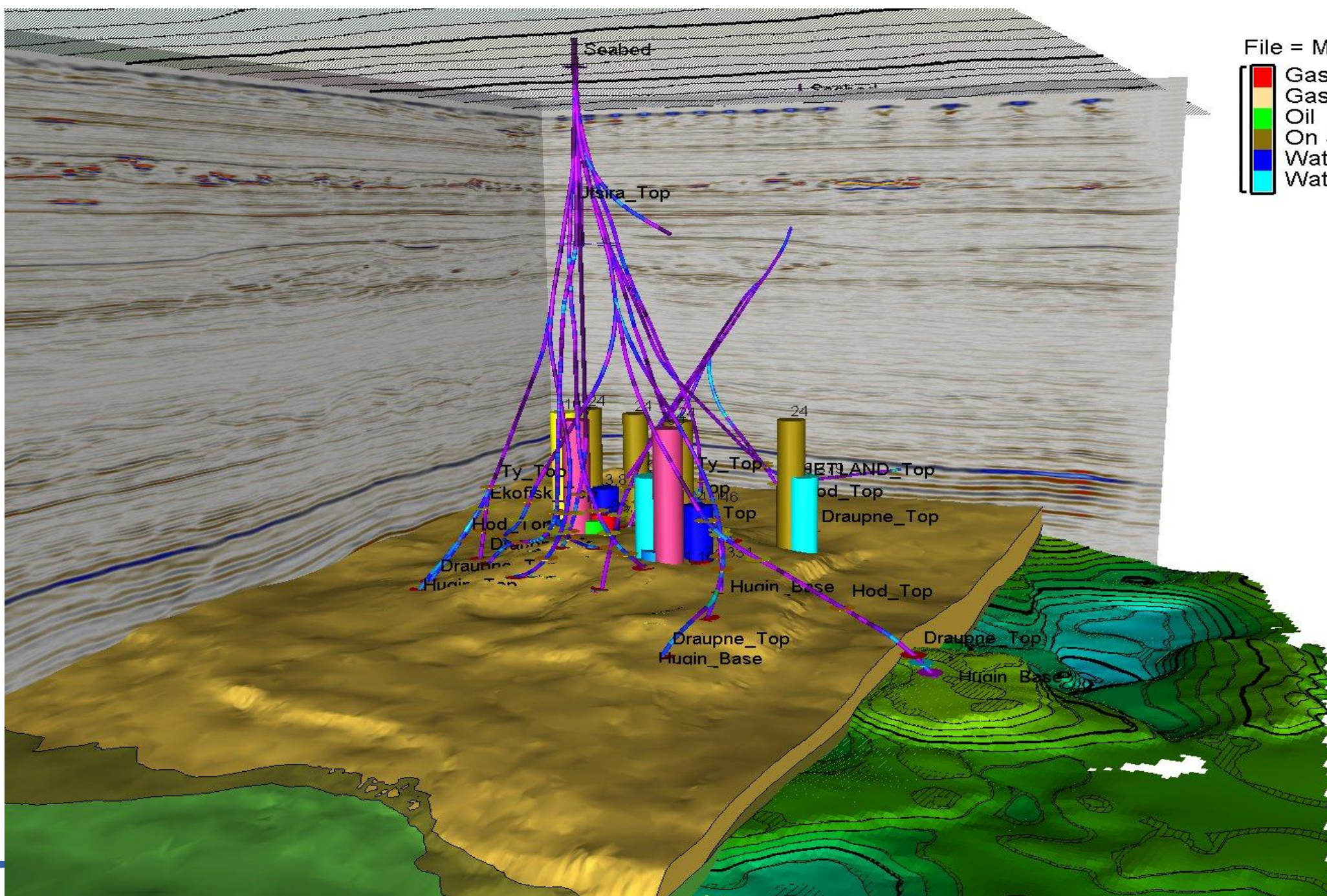
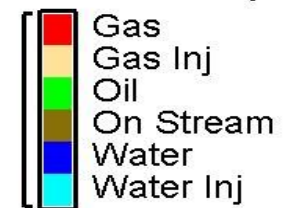
Slant (J) Type



"S" Type



Horizontal Type



Exercises

You are a well planner tasked with drilling to a reservoir located 3,000 feet east and 8,000 feet deep (True Vertical Depth). You cannot drill straight down because there's a protected forest area on the surface.

- What type of well path you'll use and why?
(Choose: Slant/J-Type, S-Type, or Horizontal.)
- What is the average inclination (angle from vertical) you'll need?

Use this formula:

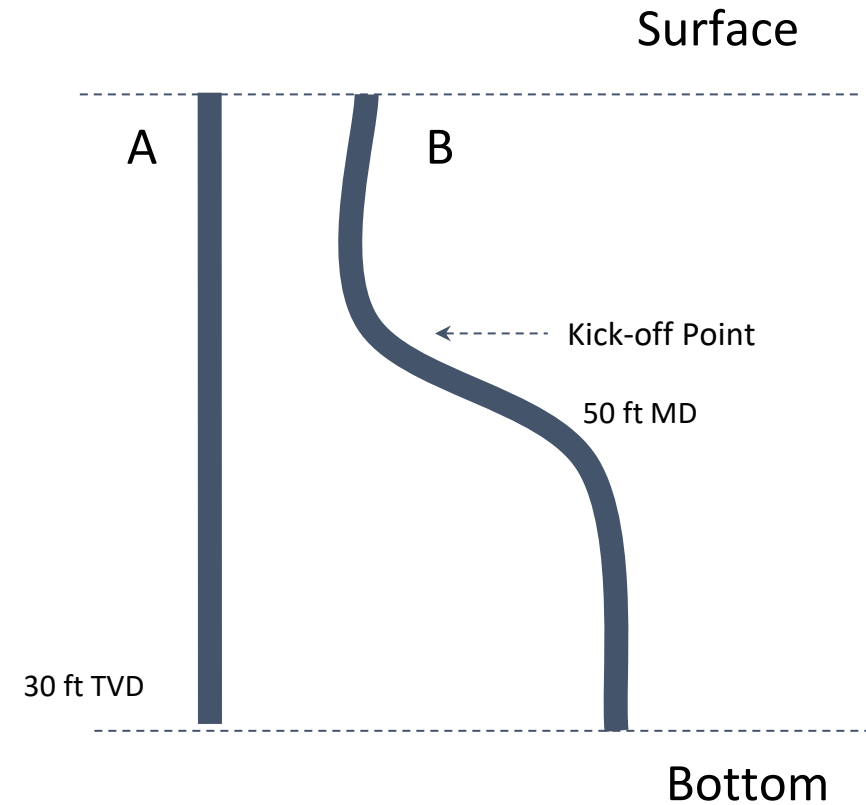
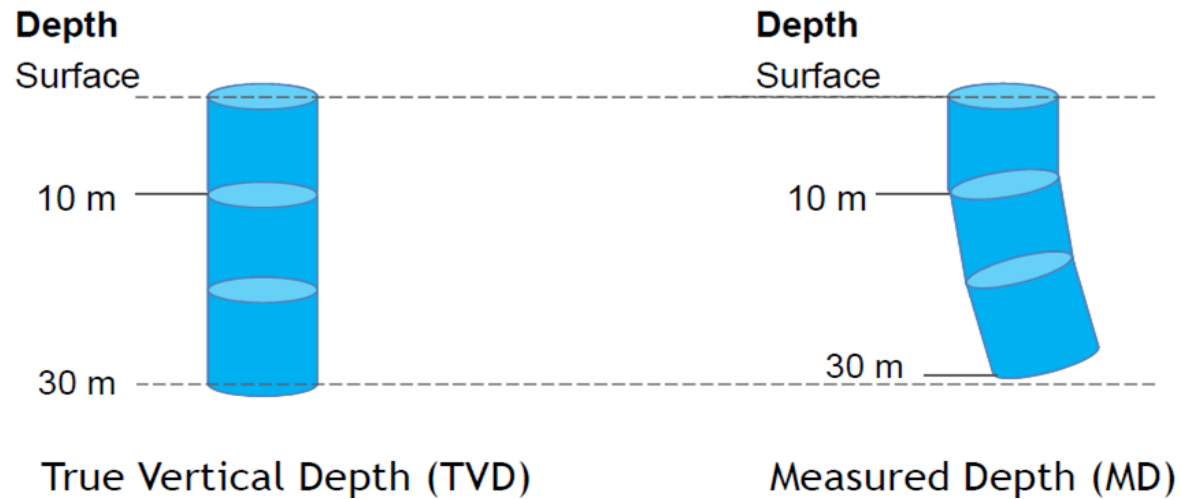
$$\text{Inclination } (^{\circ}) = \arctan \left(\frac{\text{Horizontal Distance}}{\text{TVD}} \right)$$

Well Depth

Depth is the **distance from the surface to the bottom** of the well.

Depth is used to plan drilling operations, determine casing placement, and ensure well safety.

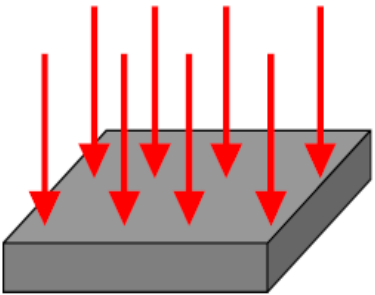
Measuring Well Depth



- Pressure = Force exerted per unit Area

$$P = \frac{F}{A}$$

Force in pounds



1 unit area (square inch)

API Units:

Pounds
Square Inch

(Pounds per Square Inch or 'psi')

SI Units:

Newton
Square Meter

(N/m² or 'Pascal')

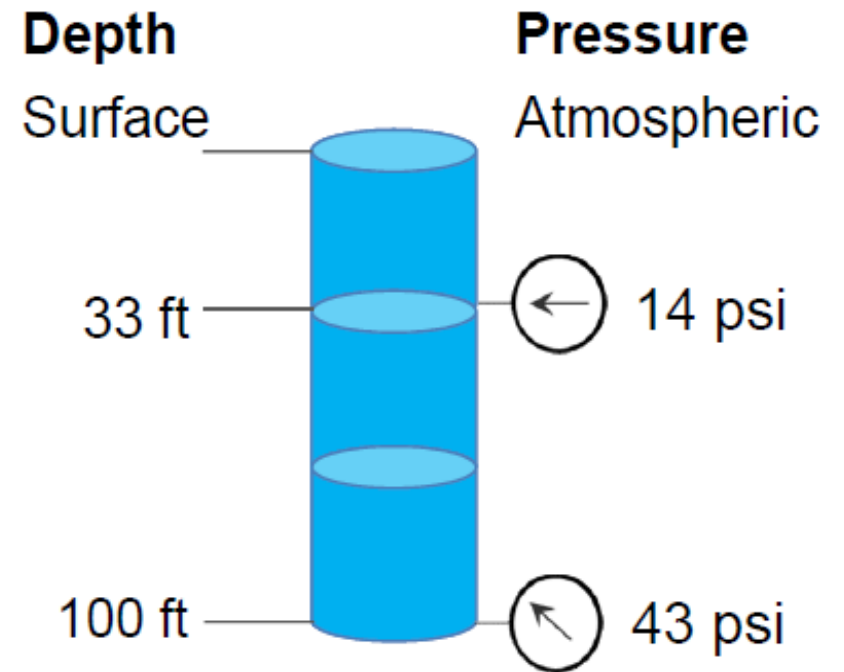
Well Depth

$\text{Mud weight (ppg)} \times 0.052 = \text{Mud Gradient (psi/ft)}$

$\text{Hydrostatic Pressure (psi)} = \text{Mud Weight (ppg)} \times 0.052 \times \text{TVD (ft)}$

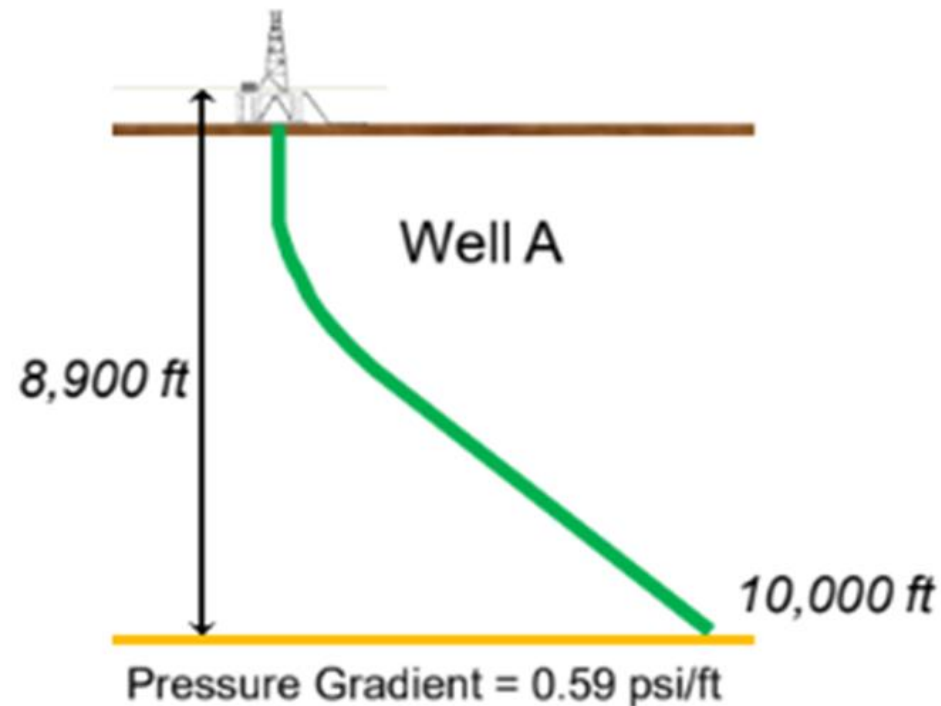
$\text{Hydrostatic Pressure (psi)} = \text{Mud gradient (psi/ft)} \times \text{TVD (ft)}$

Single Fluid Density Water (8.33 ppg)



What is the Hydrostatic Pressure at the bottom of Well A ?

$$\begin{aligned}\text{HP} &= \text{Gradient} \times \text{TVD} \\ &= 0.59 \text{ psi/ft} \times 8,900 \text{ ft} \\ &= 5251 \text{ psi}\end{aligned}$$



- Formation Pressure or **Pore Pressure** is resultant pressure of geomechanics force and fluid pressure exerting wellbore
- Formation **Fracture Pressure** is the pressure value where the formation can initiate to breakdown or crack
- Drilling fluid mud weight must be between Pore Pressure and Fracture Pressure

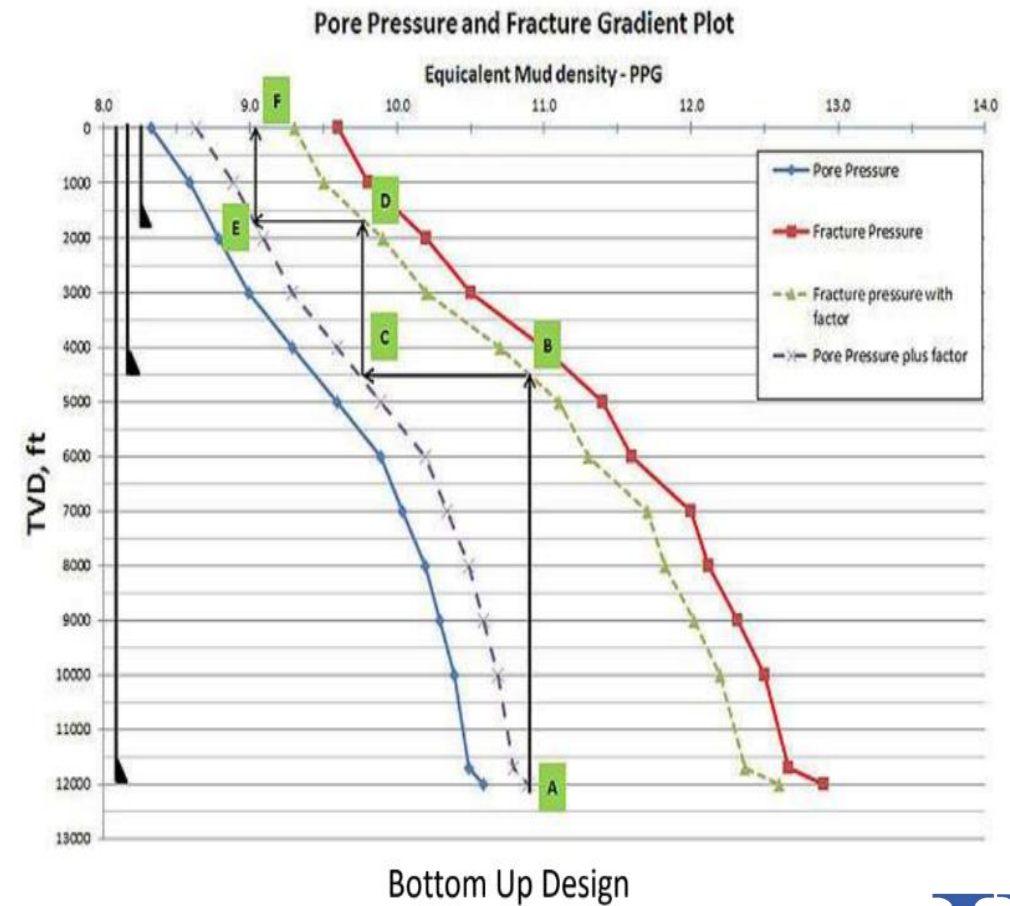
Formation Pressure

<

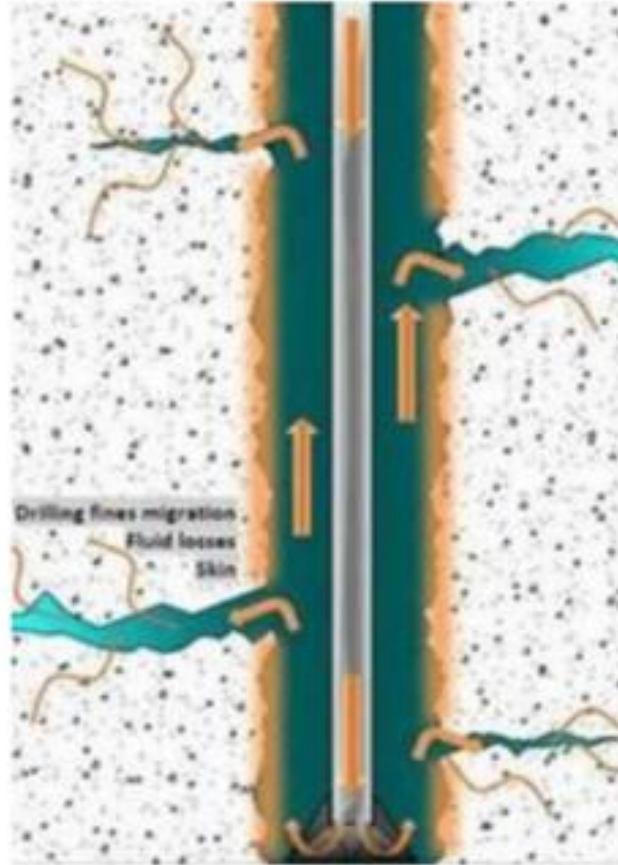
Mud Hydrostatic

<

Fracture Pressure

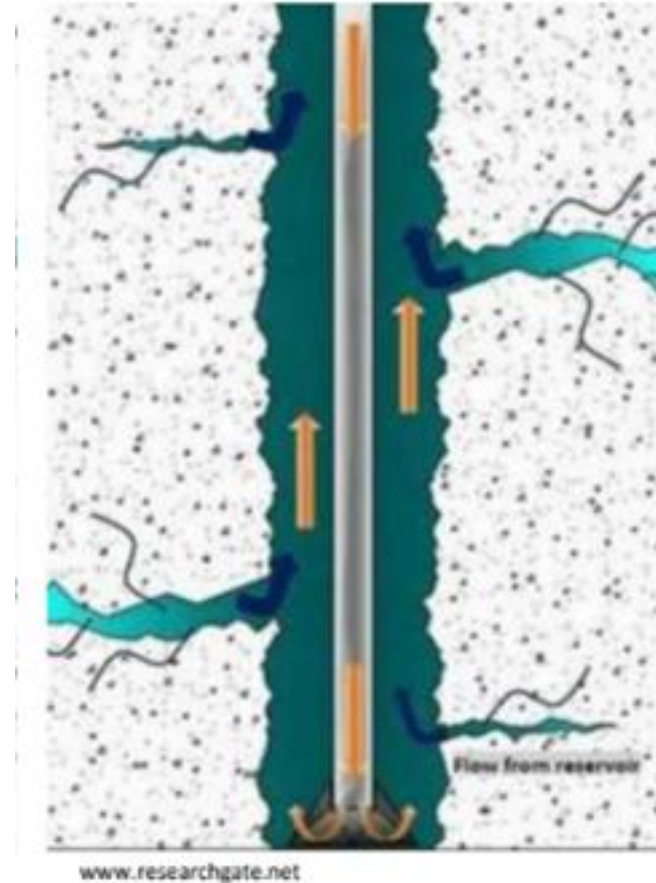


>> Overbalanced



Mud Hydrostatic $>$ Formation Pressure

<< Underbalanced



Mud Hydrostatic $<$ Formation Pressure

Well Size

Well Size is the diameter of the wellbore or hole drilled in the formation.

Well size determines the casing size, drilling tools, mud circulation, and completion design to ensure safe, efficient, and successful drilling operations.

Well Size

Conductor

- First Casing installed
- Used to mitigate broaching
- Support axial load for the entire well
- May be used to install diverter
- May be used to isolate fresh water aquifers

Surface Casing

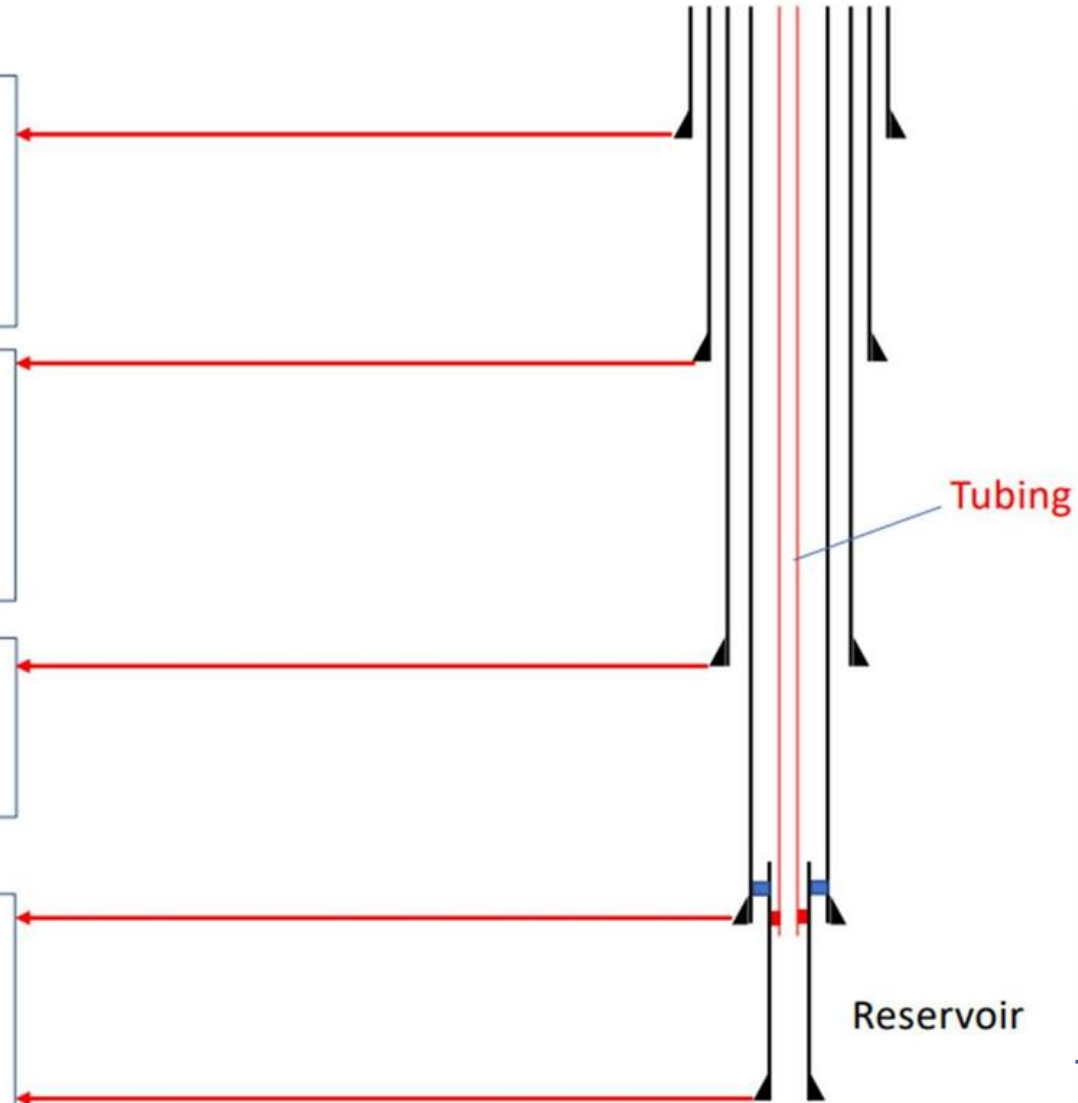
- Most important casing string to protect people and environment at surface.
- **By definition: First casing with BOP installed**
- Set at sufficient depth to mitigate broaching from a well control event
- May be used to isolate fresh water aquifers
- May be designed for compressional structural load

Intermediate Casing

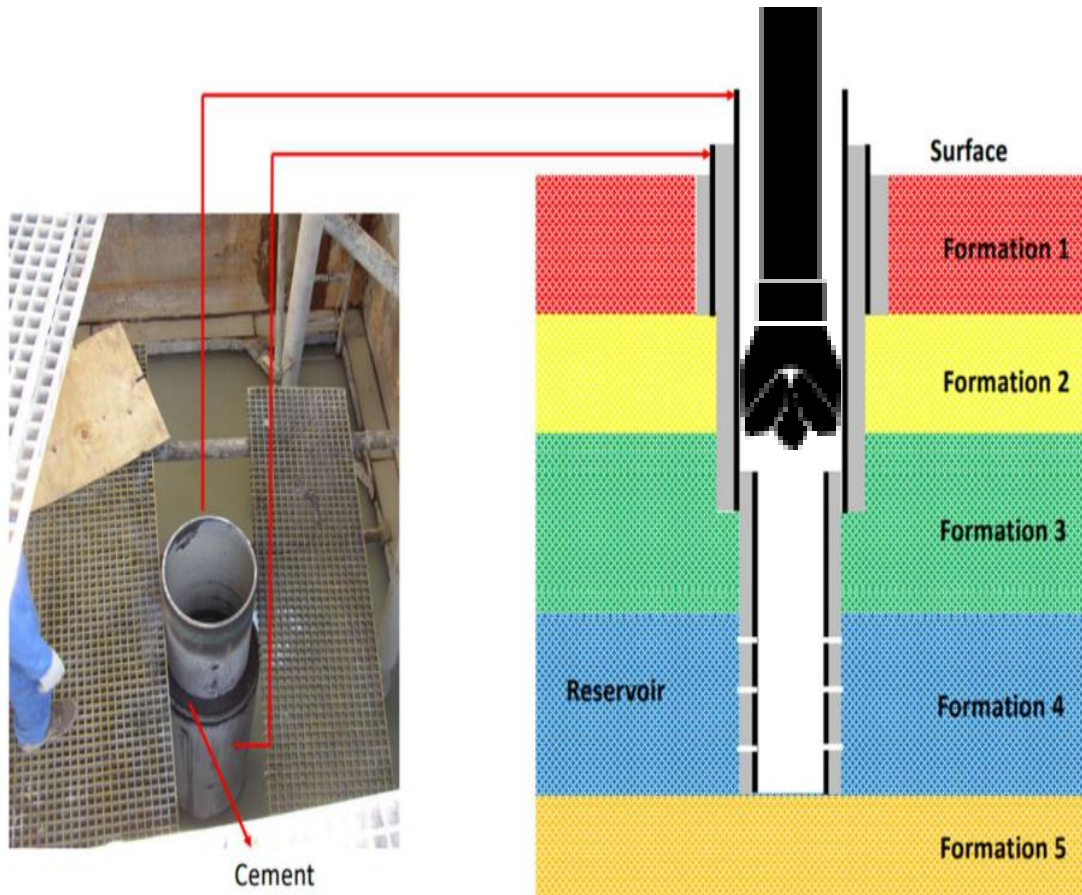
- Used to isolate formation with different pore pressure or frac pressure
- Used to isolate trouble zones (e.g. loss circulation zones, unstable formation, etc.)
- One or more strings as needed

Production casing / Production liner

- By definition, a string of casing (lower part may be a liner) that extends the entire length of the well
- Set to isolate the entire wellbore
- Used to facilitate completion parts / completion operation

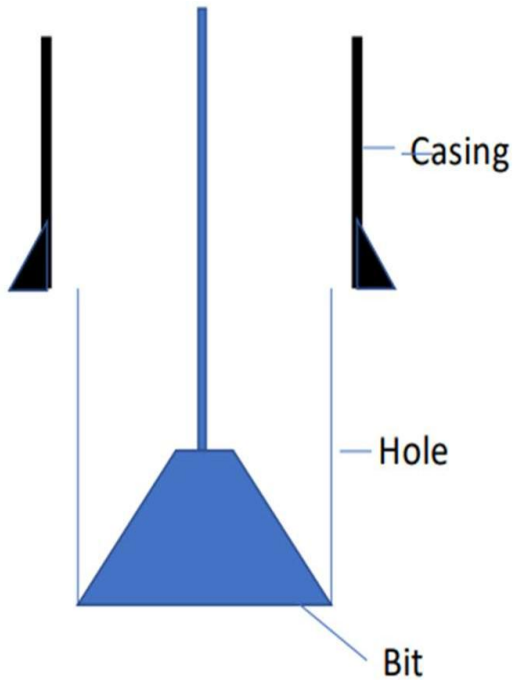


Well Size

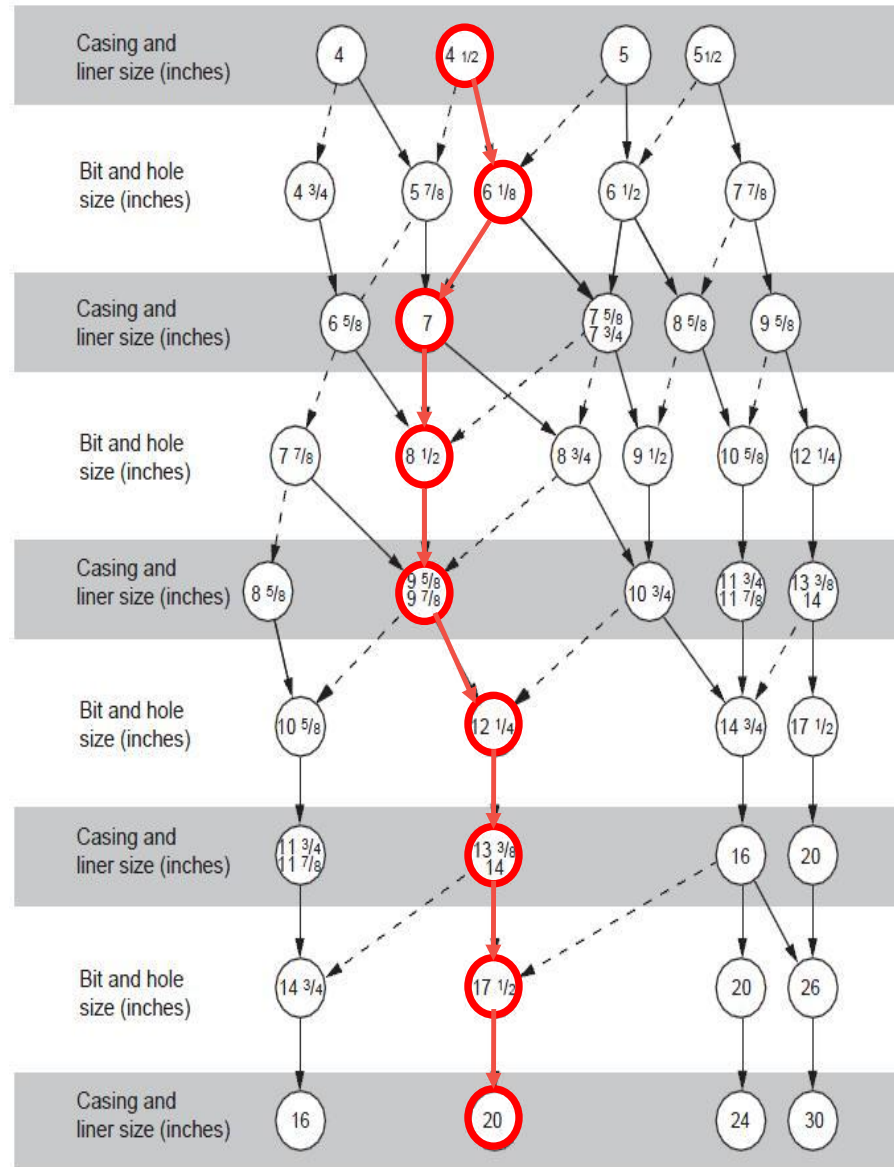


- ▶ To enable the production casing to be placed in the well, the bit size used to drill the last interval of the well must be slightly larger than the OD of the casing connectors.
- ▶ The selected bit size should provide sufficient clearance beyond the OD of the coupling to allow for mud cake on the borehole wall and for casing appliances, such as centralizers and scratchers. The bit used to drill the lower portion of the well also must fit inside the casing string above.

Well Size

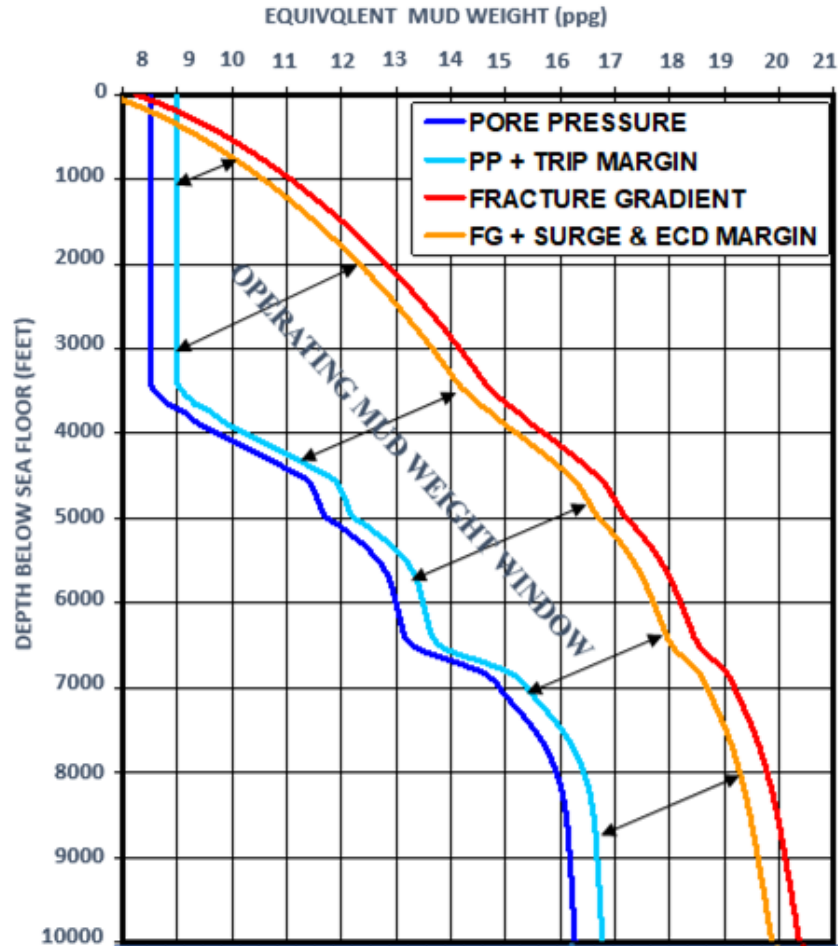


- Bit size = hole size
- Bit must be able to pass the casing. Bit size < casing drift ID



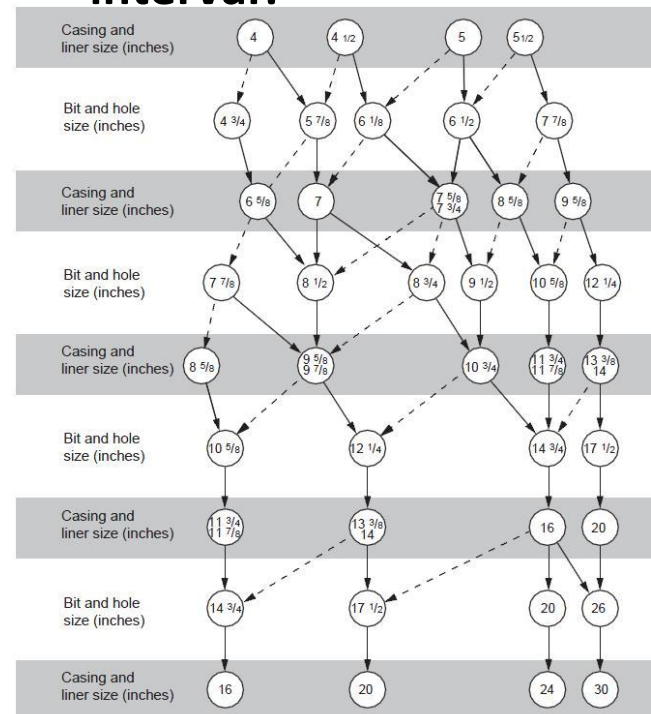
- Diameter selection from bottom of the well to the top of the well
- Size across reservoir comes from reservoir/production requirement. It may be casing or hole size (if open hole completion)

Exercises



- Proposed TD 8500 FT
- Production Casing 9 5/8

1. What is the casing setting depth for the surface casing and the production casing?
2. What casing size is appropriate for each interval?
3. What drill bit size should be used to drill each casing interval?



bit.ly/casing_exercise

WELL CONSTRUCTION PROCESS

WELL DESIGN

Well Trajectory

Well Depth

Well Size

WELL COMPONENTS

Casing Design

Wellhead Design

Cement Design

WELL EXECUTION

Drilling Rig

Drilling Tools

Drilling Fluid

WELL COMPONENTS



Casing Design

Casing is a steel pipe installed inside the well

Multiple layers of steel that protect the well from surface to reservoir

CASING STRENGTH – DESIGN CONCEPT

GOAL : how to select your casing, that is **sufficient**, but **not overly** designed.

$$\text{Casing Strength} \geq \text{Casing Load}$$

Grade,
Weight,
Size

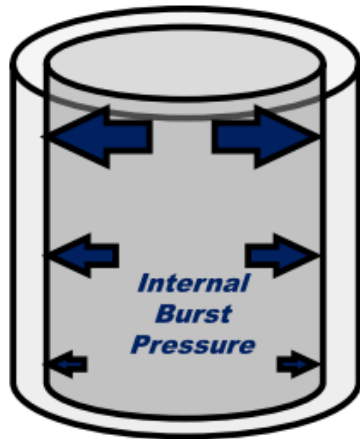


$$\text{Burst Strength} \geq \text{Burst Load}$$

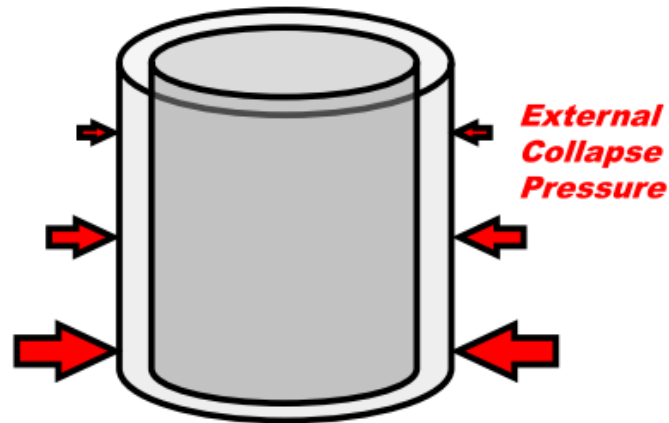
$$\text{Collapse Strength} \geq \text{Collapse Load}$$

$$\text{Axial Strength} \geq \text{Axial Load}$$

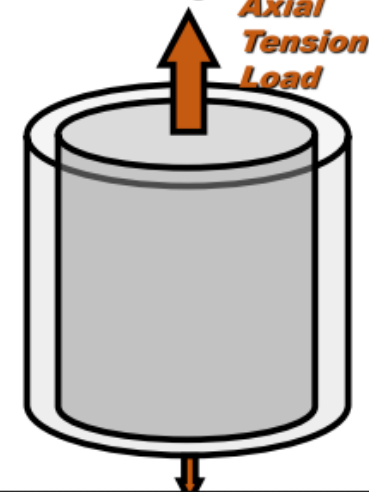
Burst Strength



Collapse Strength



Axial Strength



Casing Design

Casing is described using the following parameter (up to 20 "):

- Outer Diameter (1)
- Weight per unit length (2)
- Grade of steel (3 & 4)
- Chemical composition (3)
- Yield strength (4) → in thousands. E.g: 55 means 55,000 psi
- Length range (5)
- Type of coupling/connection (6)

18-5/8", 115 ppf, K-55, R-3, BTC

1 2 3 4 5 6



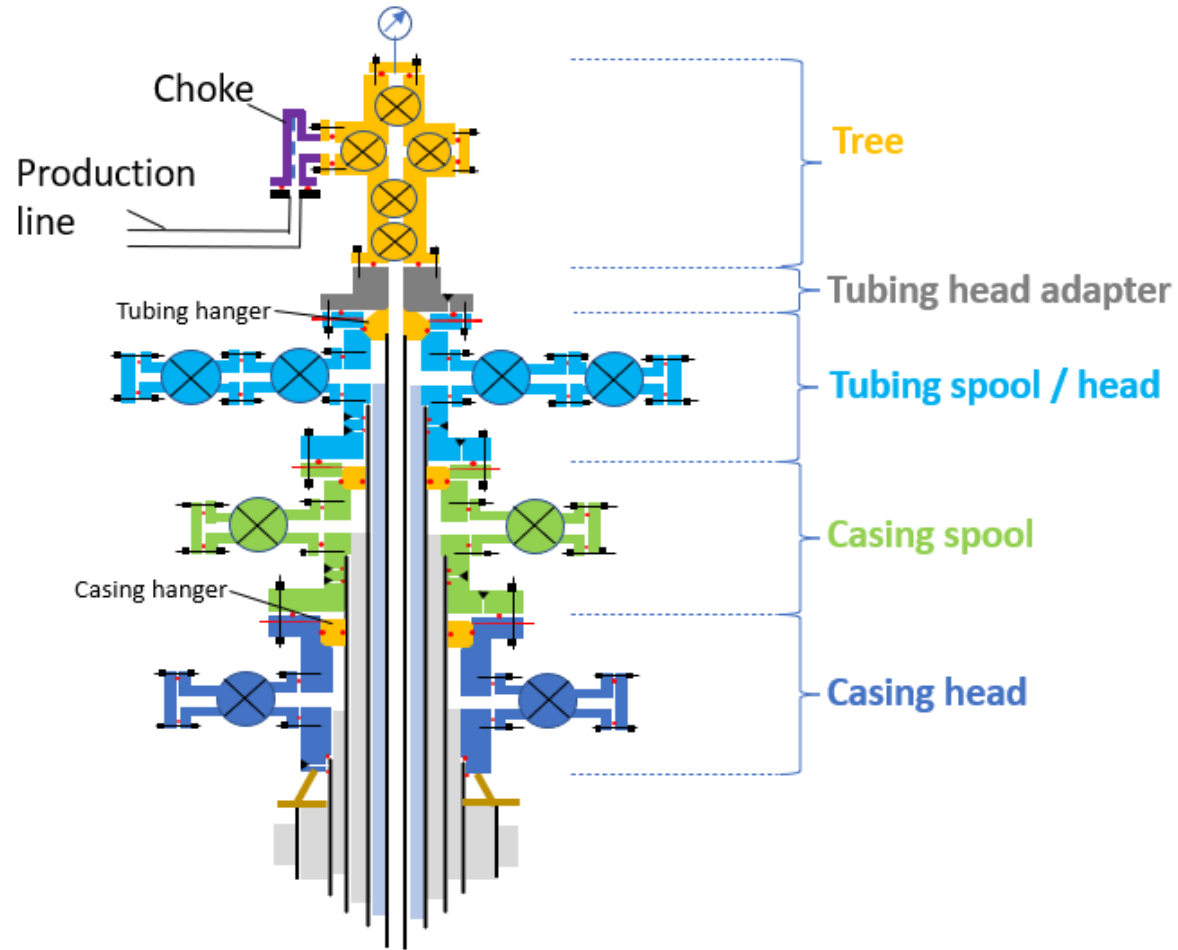
Wellhead Design

The wellhead is the equipment installed at the surface, **on top of the well**.

Its job is to **hold the casing or tubing** and **seal the well** for safe pressure control.

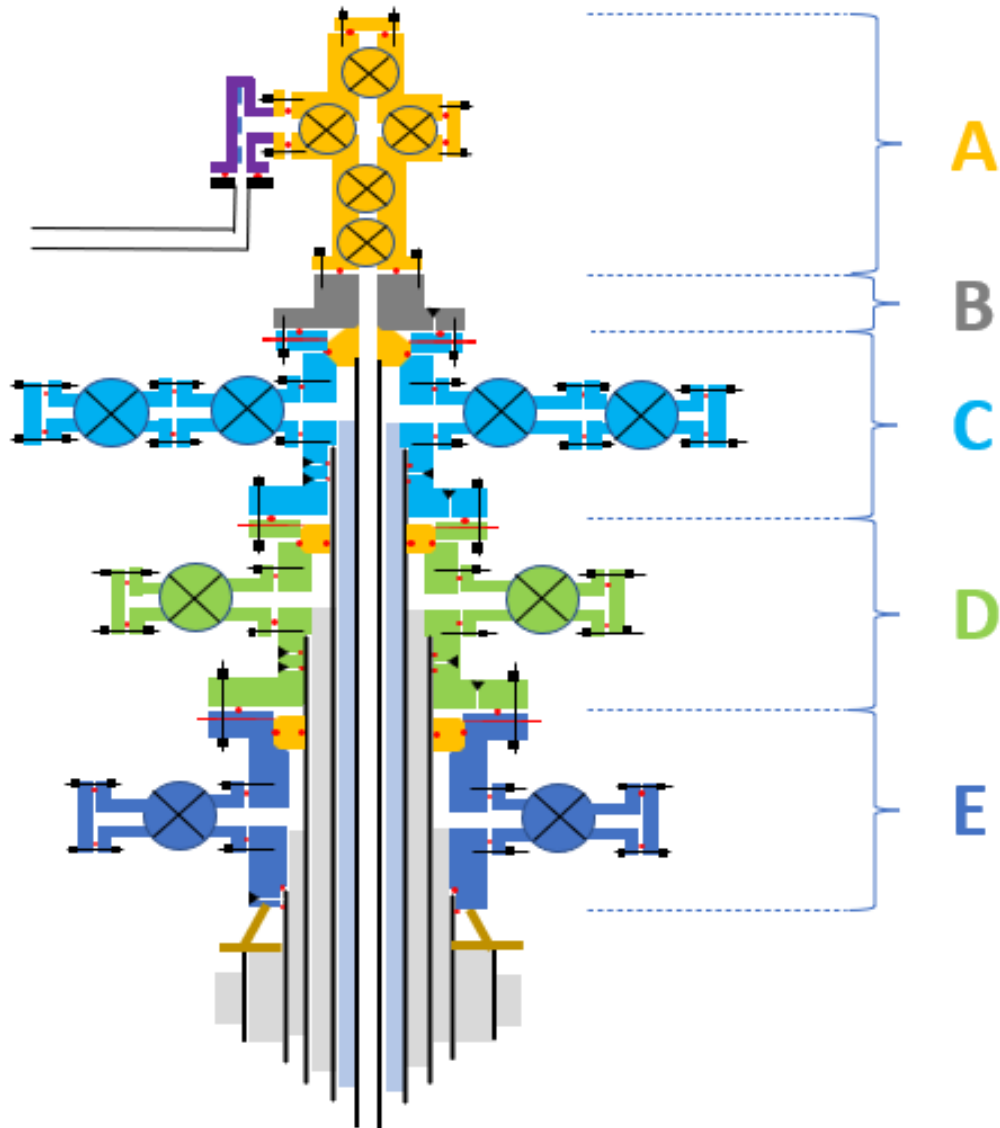


Wellhead Design



Parts	Functions
Casing Head	Welded to the top of the casing, providing a strong base for everything above.
Casing Spool	Provides side access to monitor/pump fluids between pipes.
Tubing Spool / Head	Holds the production tubing that carries oil/gas up.
Tubing Head Adapter	Connects the tubing spool to the tree.
Tree	The control panel of valves and gauges to manage well flow.
Choke	The adjustable valve that controls pressure and flow rate.
Production Line	The pipeline taking oil/gas to storage or processing.

Wellhead Design



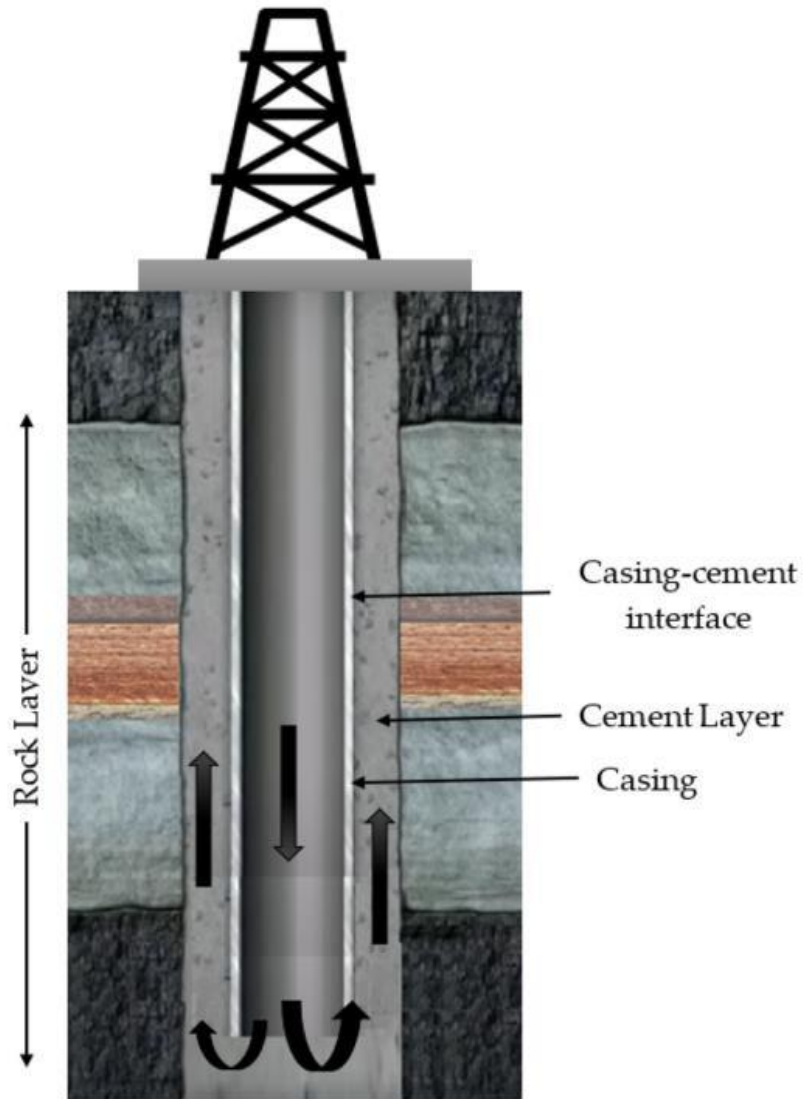
Let's Exercise!

1. Which part is welded directly to the top of the casing?
1. Which part holds the smaller pipe (tubing) that carries oil/gas to the surface?
1. Which part is the “control panel” with valves and gauges to manage the well?

Cement Design

Cementing is the process of pumping cement slurry into the well to **fill the space between the casing and the rock formation.**

Its main purpose is to **permanently seal and secure the casing in place.**



How Cementing Works?

Step 1 – Run Casing

The steel casing pipe is lowered into the drilled hole.

Step 2 – Pump Cement

A special cement slurry is pumped down the casing and out the bottom.

It flows back up outside the casing, filling the annular space.

Step 3 – Displace with Mud

Drilling mud is pumped behind the cement to push it fully into place.

Plugs separate the cement from the mud to keep them from mixing.

Step 4 – Let It Harden (WOC)

We Wait On Cement (WOC) for it to harden (like concrete).

Once set, the casing is permanently sealed and supported.

WELL CONSTRUCTION PROCESS

WELL DESIGN

Well Trajectory

Well Depth

Well Size

WELL COMPONENTS

Casing Design

Wellhead Design

Cement Design

WELL EXECUTION

Drilling Rig

Drilling Tools

Drilling Fluid

WELL EXECUTION

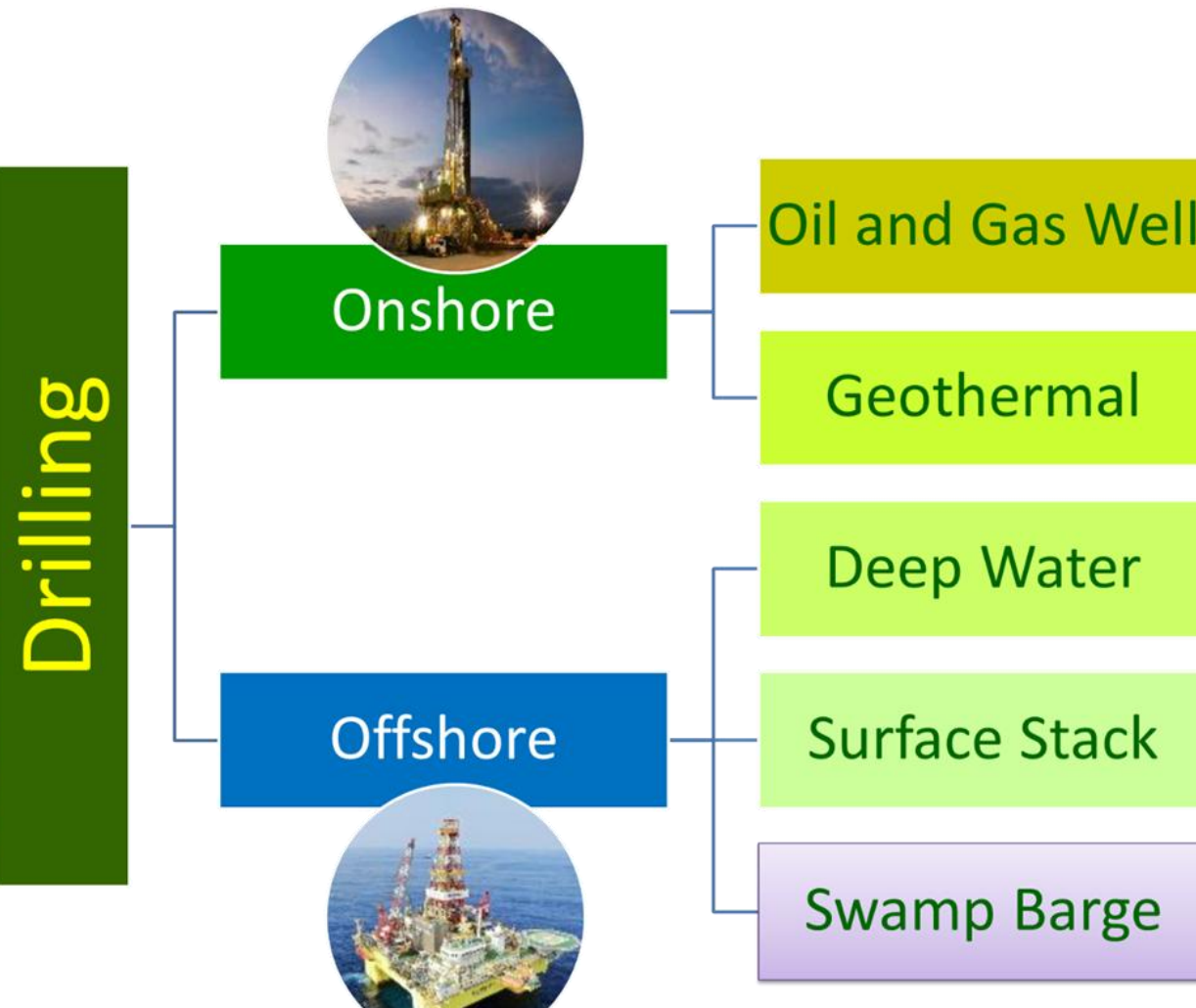


Drilling Rig

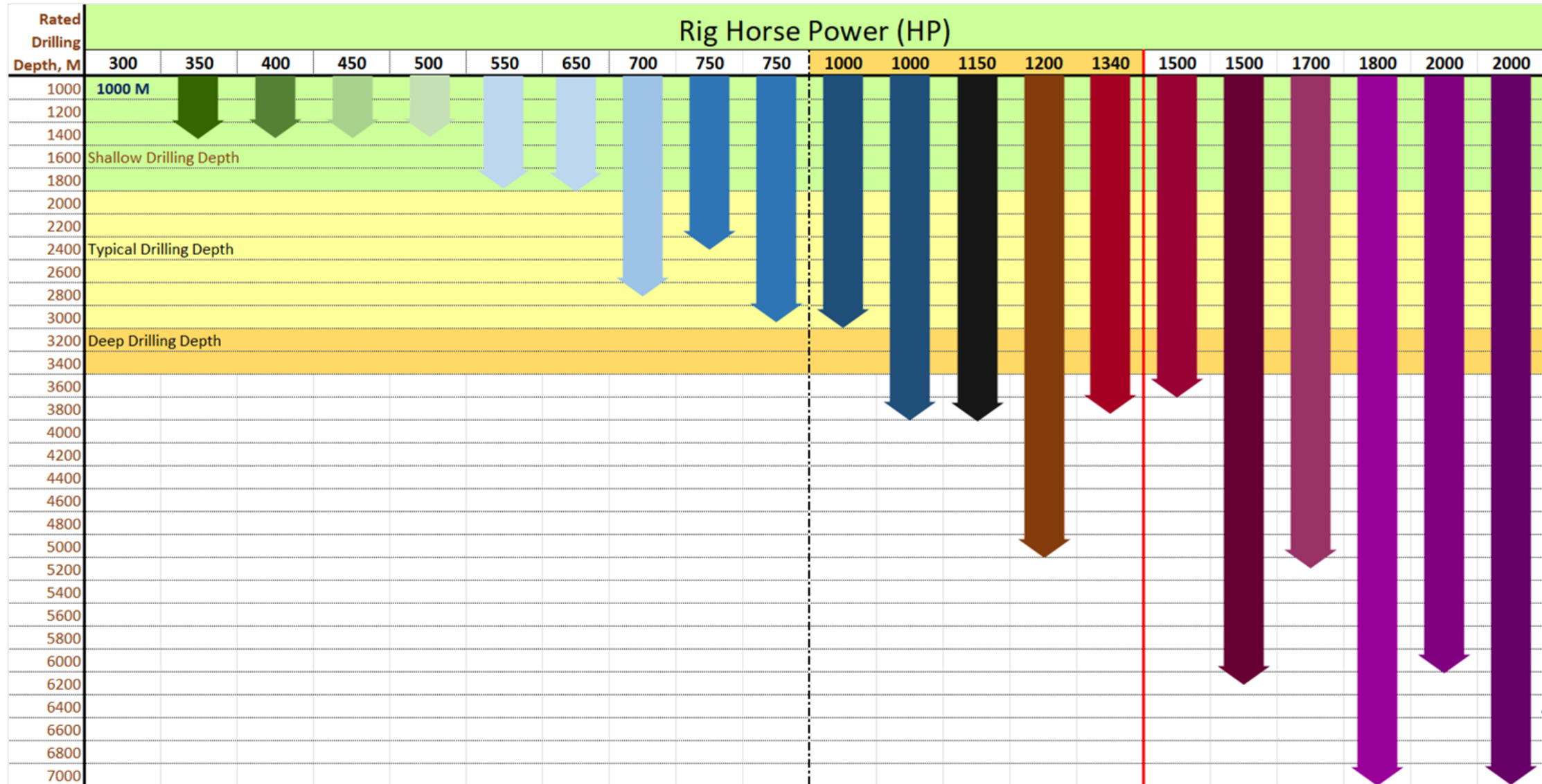
Drilling Rig is a set of equipments that Drills the Well

Its function is to drill a hole into the earth to reach Oil and Gas resources.

DRILLING Categories Based On LOCATION



Drilling Rig



Oil and Gas Drilling Work Location



Geothermal Drilling Work Location



1. Power system

Required for rotating, hoisting, fluid circulation, mud cleaning and auxiliary equipment, BOP

2. Hoisting system

Running and pulling the drilling assembly and other related surface and subsurface equipment

3. Rotary system

top drive

4. Circulating system

Drilling fluid circulating system

5. Well control system

BOP system

RIG POWER SYSTEM

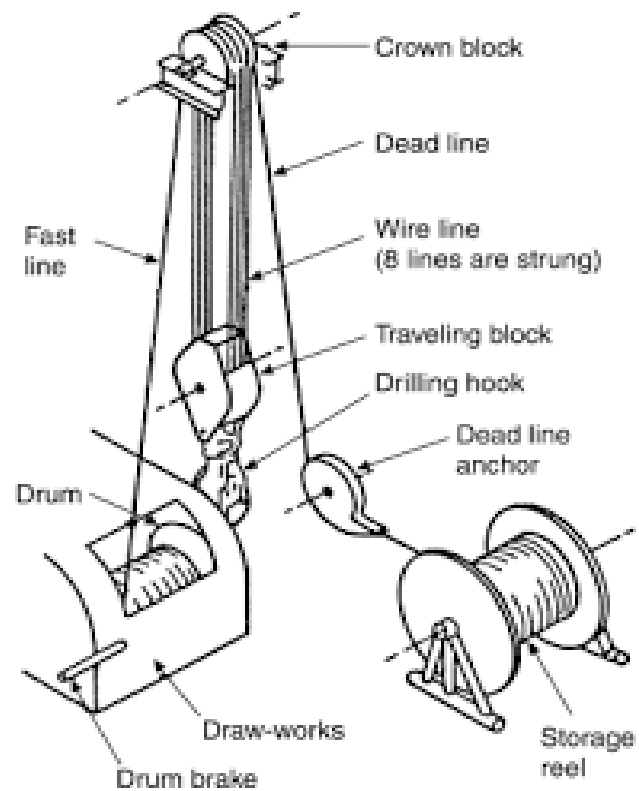
Rig Power Requirements:

- Power for hoisting
- Power for rotating the drill string
- Hydraulic power for circulating
- Power for auxiliary equipment

Most of the rigs today are diesel powered.



Hoisting system



Rotary System

Top Drive System (Electrical & Hydraulic)



Split Block

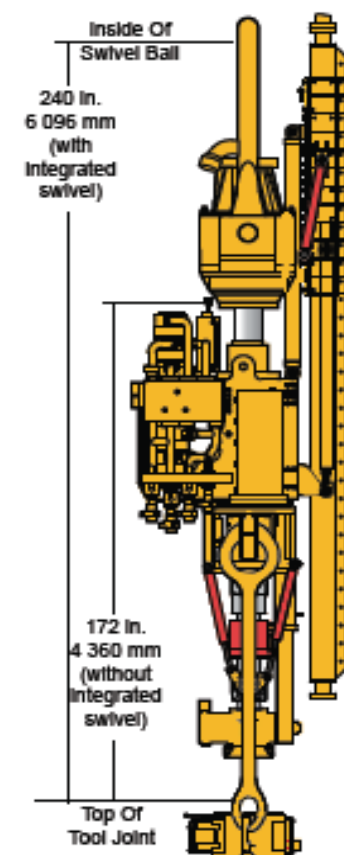
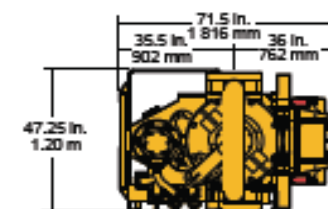
Swivel with space for Wireline BOP

Top Drive System

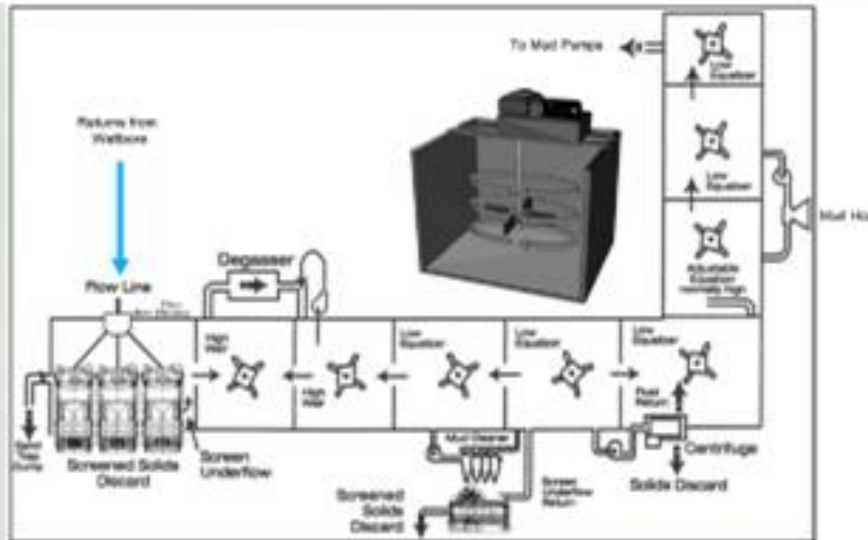
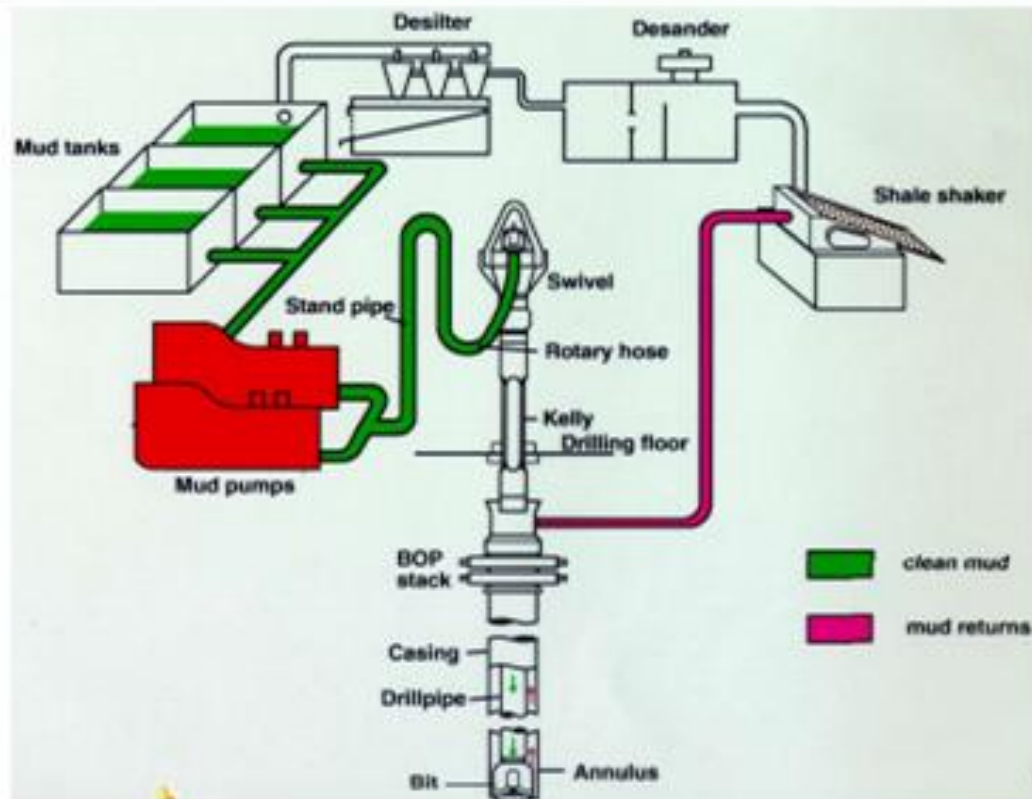
Casing Drive System

10 3/4-in. casing

ConocoPhillips



Circulating System



Courtesy of Derrick Equipment

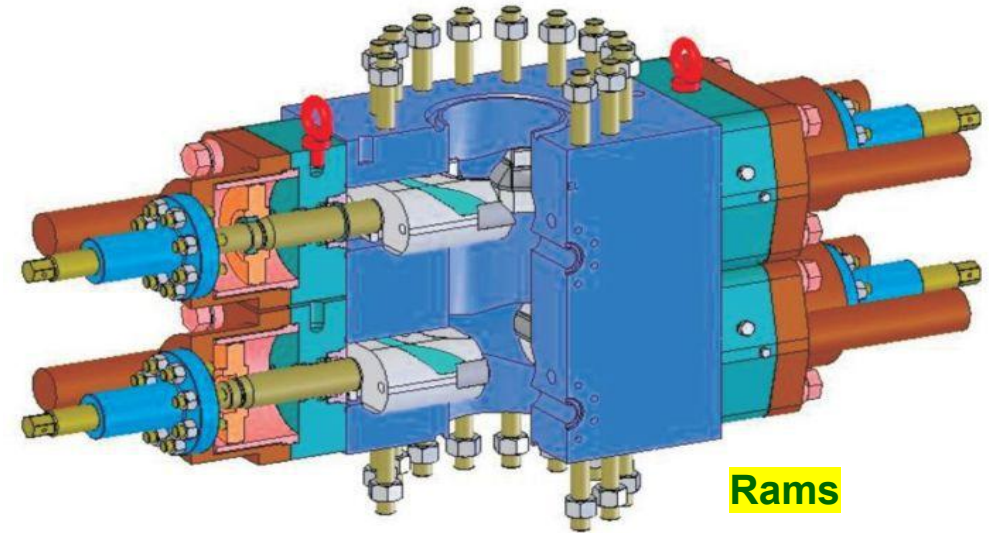
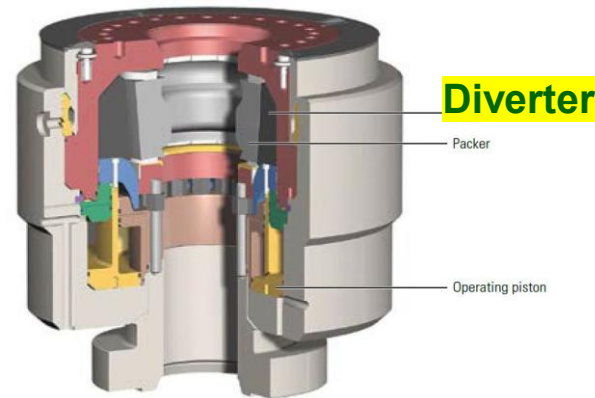
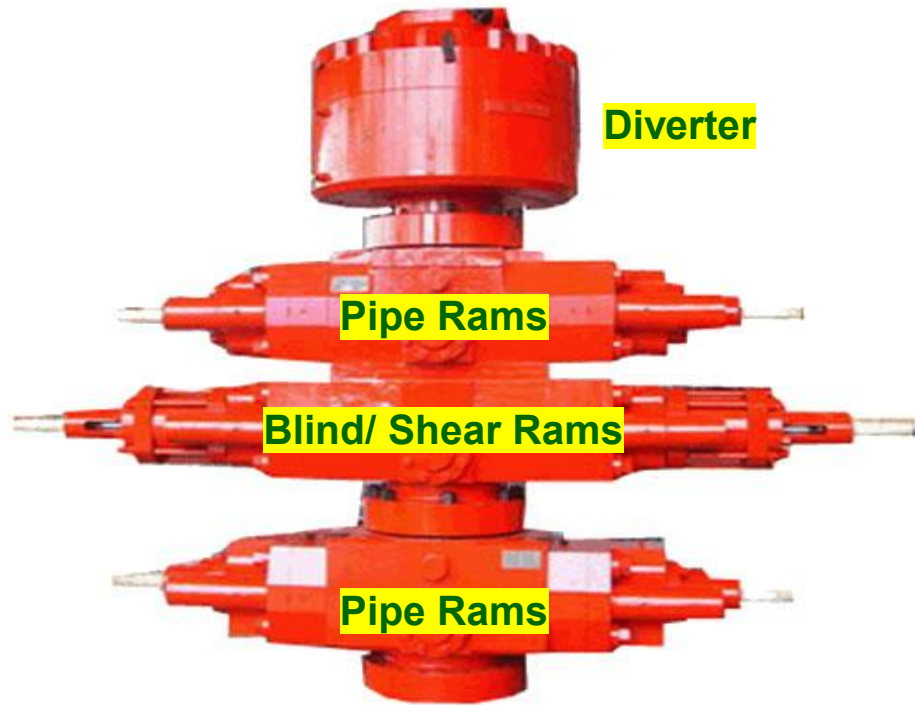


<https://www.grsolidcontrol.com/mud-tank>



<https://www.now.com/products/rapid-mud-tank-system>

Well Control System



The ***Blowout Prevention System*** on a drilling rig is the system that prevents the uncontrolled, catastrophic release of high-pressure fluids (oil, gas, or salt water) from subsurface formations.

Drilling Tools

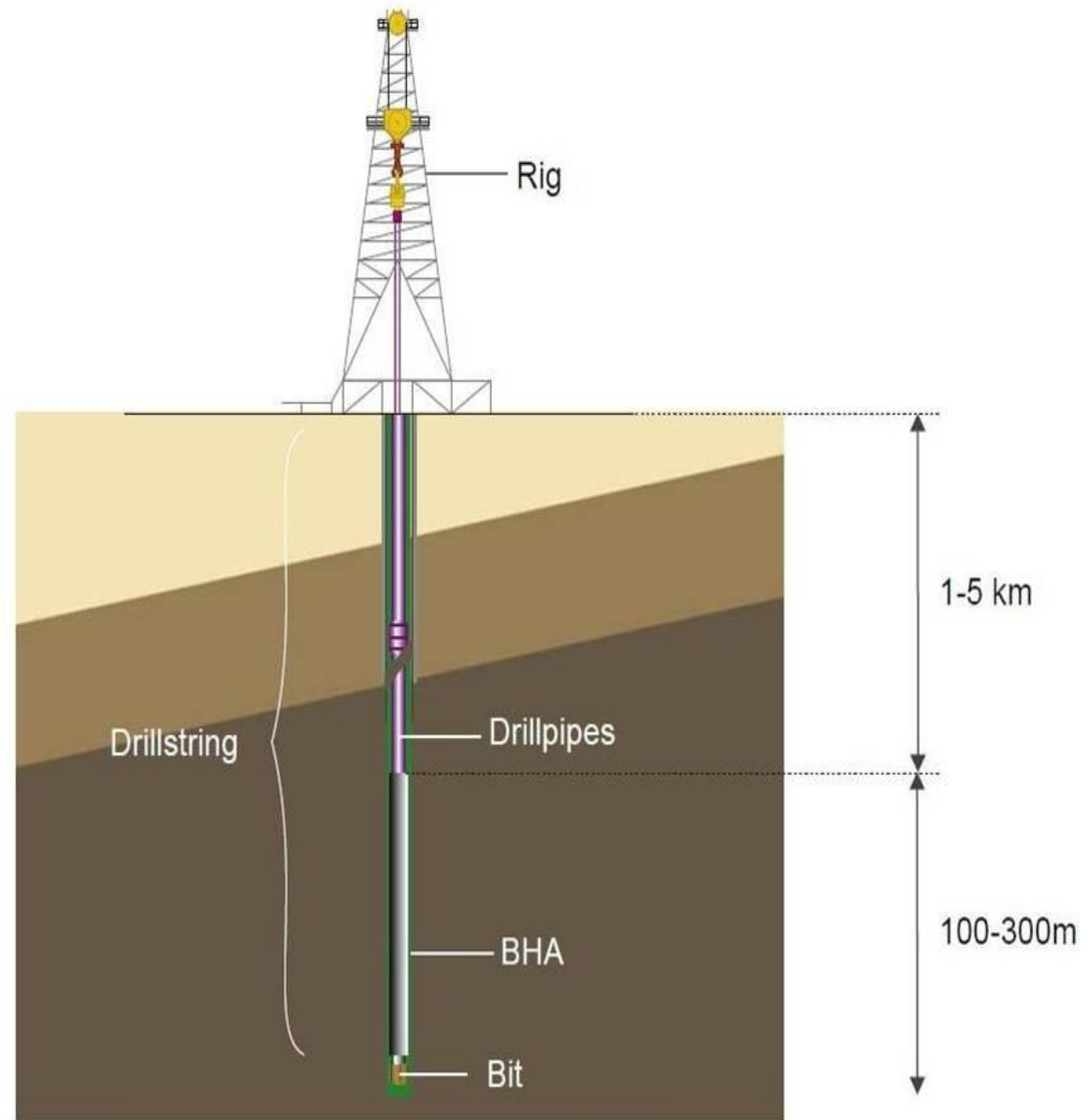
Drilling Tools are the equipment used in drilling operations to create the wellbore and support the drilling process.

Drilling tools are used to drill the hole, transmit rotation and weight to the drill bit, circulate drilling mud, and ensure safe and efficient well construction.

Drilling Tools

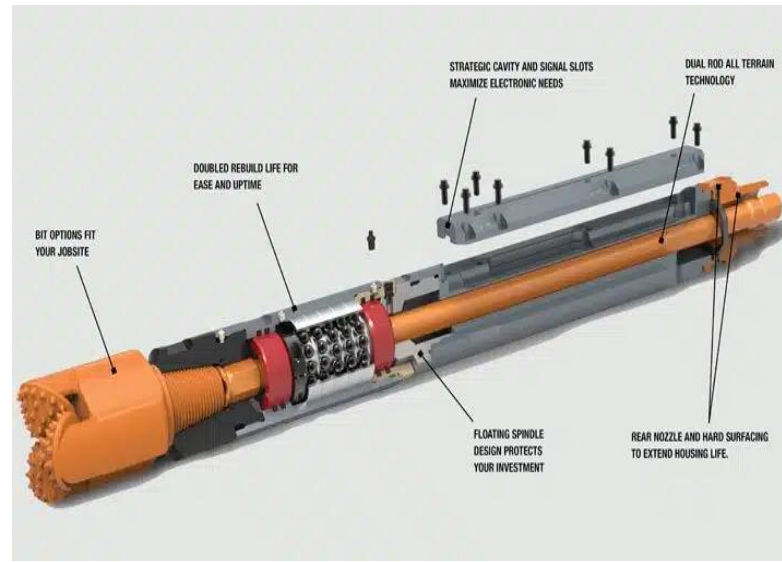
- Drilling Bits
- Mud Motor
- MWD / LWD
- Stabilizer
- Drill Collar
- HWDP
- Drill Pipe

BHA

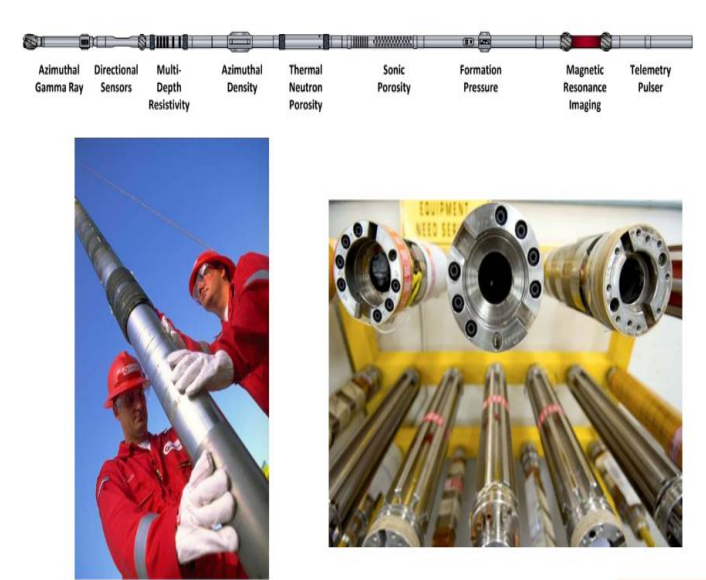




Drilling Bits



Mud Motor



LWD



Stabilizer



Drill Collar



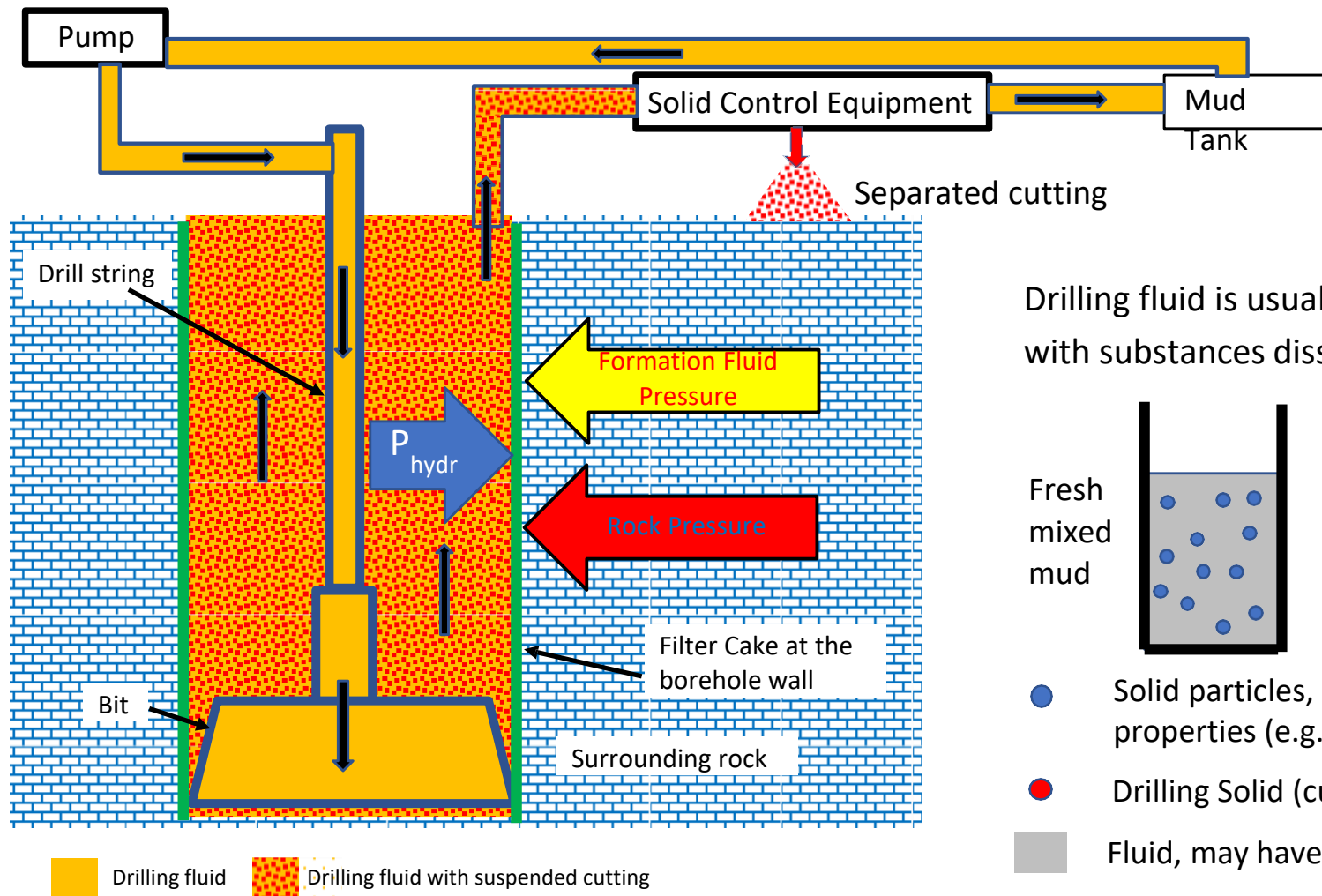
HWDP

Drilling Fluid

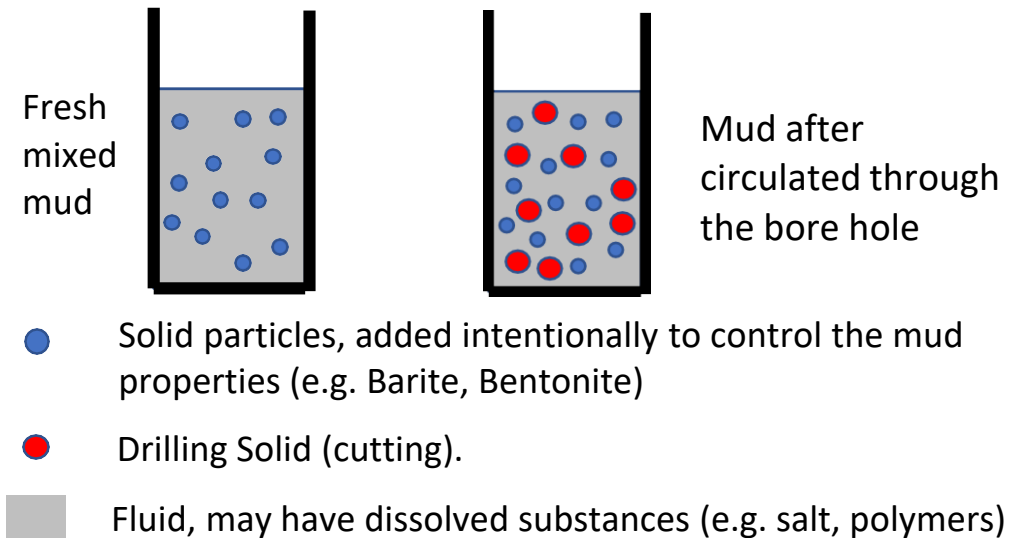
Drilling Fluid (also called drilling mud) is a fluid used during drilling operations to circulate inside the well.

It is used to **cool and lubricate the drill bit, carry drill cuttings to the surface, control formation pressure, and maintain wellbore stability.**

Drilling Fluid



Drilling fluid is usually a mixture of a liquid (water or oil), with substances dissolved, and solids suspended in it.



January, 2026

JDS
Jakarta Drilling Society



JAKARTA DRILLING SOCIETY

THANK YOU